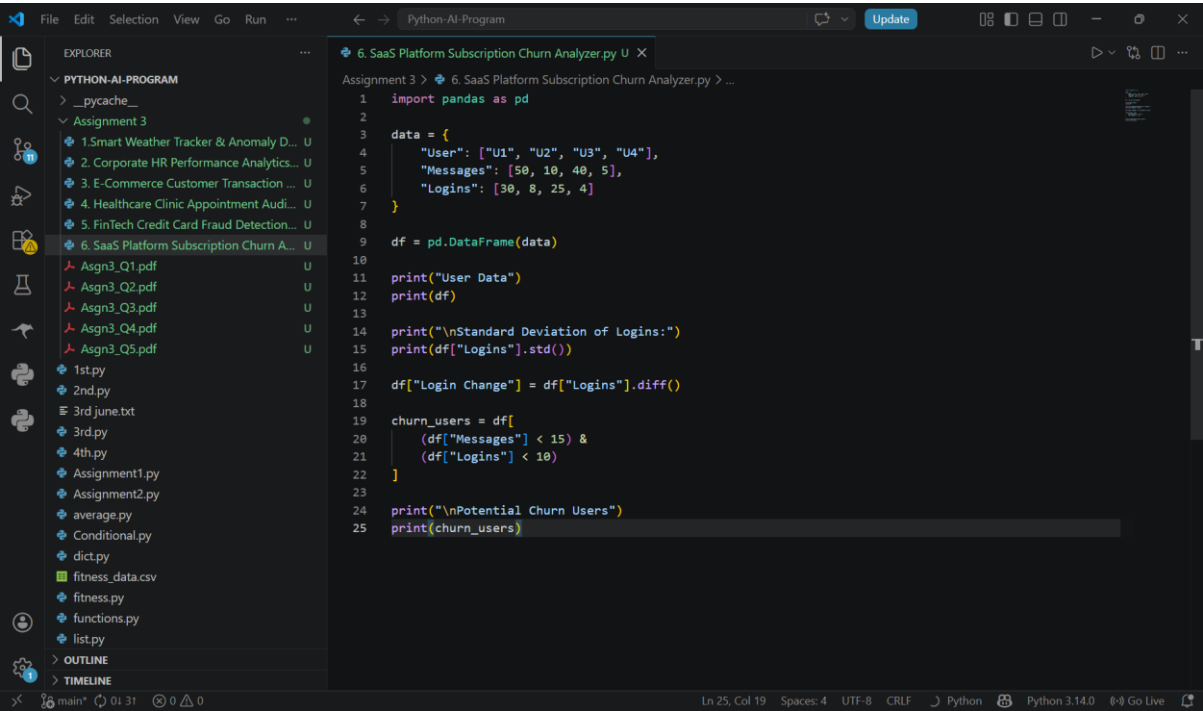
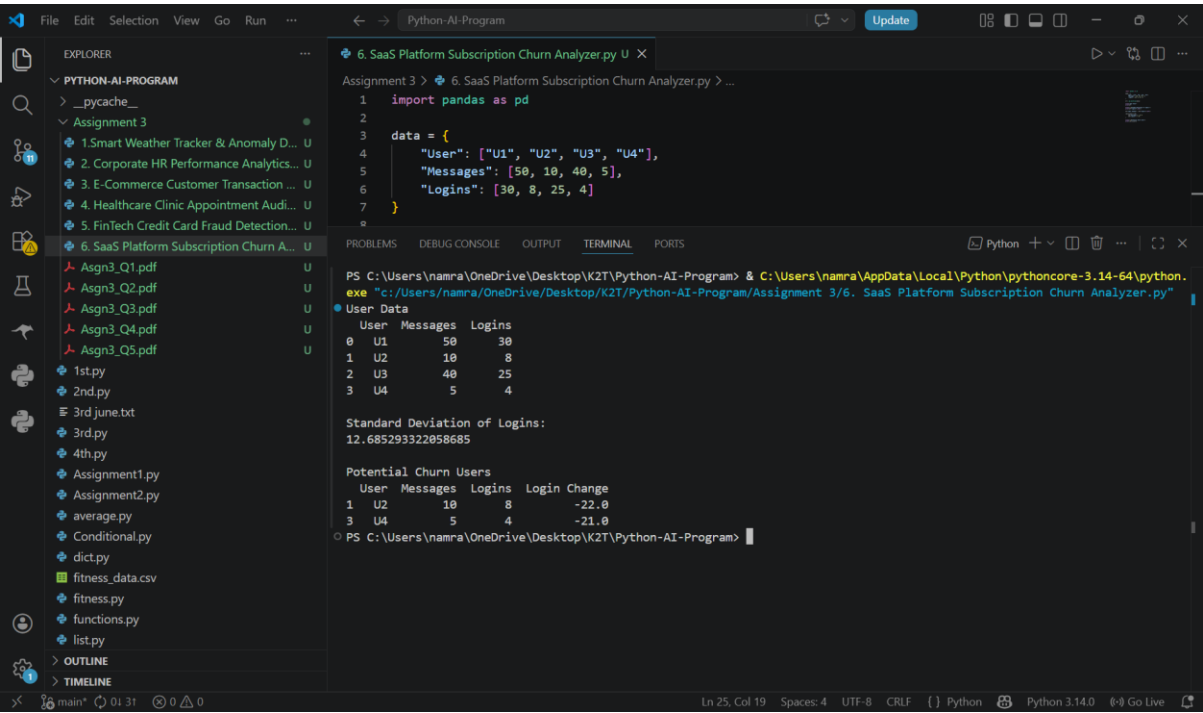


CODE: 6. SaaS Platform Subscription Churn Analyzer



```
1 import pandas as pd
2
3 data = {
4     "User": ["U1", "U2", "U3", "U4"],
5     "Messages": [50, 10, 40, 5],
6     "Logins": [30, 8, 25, 4]
7 }
8
9 df = pd.DataFrame(data)
10
11 print("User Data")
12 print(df)
13
14 print("\nStandard Deviation of Logins:")
15 print(df["Logins"].std())
16
17 df["Login Change"] = df["Logins"].diff()
18
19 churn_users = df[
20     (df["Messages"] < 15) &
21     (df["Logins"] < 10)
22 ]
23
24 print("\nPotential Churn Users")
25 print(churn_users)
```

OUTPUT:



```
PS C:\Users\namra\OneDrive\Desktop\K2T\Python-AI-Program> & C:\Users\namra\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/namra/OneDrive/Desktop/K2T/Python-AI-Program/Assignment 3/6. SaaS Platform Subscription Churn Analyzer.py"

User Data
User  Messages  Logins
0    U1        50      30
1    U2         10       8
2    U3        40     25
3    U4          5       4

Standard Deviation of Logins:
12.685293322058685

Potential Churn Users
User  Messages  Logins  Login Change
1    U2         10       8         -22.0
3    U4          5       4         -21.0
```